

TOL'KA, Russian Federation

WMO: 236620

Lat: 63.9944N

Lon: 82.0475E

Elev: 35

StdP: 100.91

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -45.3	(c) -41.9	(d) -48.1	(e) 0.0	(f) -45.5	(g) -44.8	(h) 0.0	(i) -41.9	(j) 6.2	(k) -15.5	(l) 5.4	(m) -15.3	(n) 0.6	(o) 270	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.9	(c) 29.3	(d) 18.2	(e) 26.9	(f) 17.2	(g) 24.5	(h) 16.5	(i) 19.3	(j) 26.1	(k) 18.3	(l) 24.8	(m) 17.2	(n) 23.2	(o) 2.6	(p) 150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.0	12.2	21.9	15.8	11.3	20.5	14.8	10.6	19.4	55.3	26.0	51.9	25.0	48.7	23.0	22.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 5.9	(b) 5.2	(c) 4.5	(e) -46.2	(f) 31.6	(g) 2.8	(h) 1.7	(i) -48.2	(j) 32.9	(k) -49.8	(l) 33.9	(m) -51.4	(n) 34.9	(o) -53.4	(p) 36.1		
(a) 5.9	(b) 5.2	(c) 4.5	(e) -46.2	(f) 20.7	(g) 2.8	(h) 1.2	(i) -48.2	(j) 21.6	(k) -49.8	(l) 22.3	(m) -51.4	(n) 23.0	(o) -53.4	(p) 23.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-4.5	-25.4	-22.1	-11.2	-4.2	1.7	14.1	16.1	12.2	6.3	-3.0	-17.6	-21.6	(6)
(7)		DBStd	16.08	9.06	9.31	7.86	6.75	5.59	5.60	4.53	3.76	4.49	6.72	9.77	10.87	(7)
(8)		HDD10.0	5731	1096	900	657	426	265	25	4	20	128	404	827	978	(8)
(9)		HDD18.3	8369	1354	1133	916	675	515	149	99	192	360	662	1077	1237	(9)
(10)		CDD10.0	455	0	0	0	0	8	147	193	88	18	1	0	0	(10)
(11)		CDD18.3	53	0	0	0	0	1	21	29	2	0	0	0	0	(11)
(12)		CDH23.3	722	0	0	0	0	7	286	401	28	0	0	0	0	(12)
(13)		CDH26.7	203	0	0	0	0	2	76	125	0	0	0	0	0	(13)
(14)	Wind	WSAvg	2.1	1.8	1.8	2.4	2.5	2.7	2.3	2.0	2.0	2.2	2.3	2.0	1.8	(14)
(15)	Precipitation	PrecAvg	550	28	23	26	32	44	55	68	78	56	64	42	35	(15)
(16)		PrecMax	749	52	49	62	77	90	113	142	155	125	120	76	58	(16)
(17)		PrecMin	352	6	10	6	10	16	17	7	25	17	19	18	22	(17)
(18)		PrecStd	98	12	10	12	15	18	25	35	32	22	22	14	10	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-6.0	-0.2	4.2	13.2	23.2	31.3	32.1	25.9	22.1	13.1	3.0	-0.4	(19)
(20)			MCWB	-6.3	-1.0	0.3	6.4	13.2	18.9	19.3	17.2	15.0	9.6	2.3	-1.1	(20)
(21)		2%	DB	-8.2	-4.5	2.3	9.4	18.4	29.1	29.9	23.9	18.6	9.3	0.7	-2.2	(21)
(22)			MCWB	-8.7	-5.4	-0.4	4.0	11.4	18.1	18.5	16.7	13.4	7.5	0.0	-2.7	(22)
(23)		5%	DB	-10.1	-6.4	0.8	7.0	14.2	26.8	27.9	21.6	15.8	6.8	-1.0	-4.3	(23)
(24)			MCWB	-10.5	-7.0	-1.0	3.0	9.3	16.9	17.8	15.9	12.2	5.5	-1.6	-4.8	(24)
(25)		10%	DB	-12.3	-9.0	-0.6	4.8	10.4	24.4	25.4	19.2	13.2	5.0	-3.1	-6.8	(25)
(26)			MCWB	-12.8	-9.5	-2.1	1.9	6.1	16.0	17.1	14.9	10.3	3.7	-3.6	-7.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-6.4	-1.0	0.9	6.3	14.0	20.4	21.4	19.5	15.9	10.0	2.4	-1.0	(27)
(28)			MCDWB	-6.2	-0.3	2.8	11.1	20.3	28.4	28.7	23.2	20.7	12.1	3.0	-0.4	(28)
(29)		2%	WB	-8.6	-5.3	0.0	4.5	12.1	19.0	19.8	17.8	14.0	7.6	0.2	-2.6	(29)
(30)			MCDWB	-8.1	-4.7	1.8	8.3	17.4	25.9	27.2	21.8	17.4	8.9	0.7	-2.1	(30)
(31)		5%	WB	-10.6	-7.1	-0.9	3.5	9.5	18.0	18.6	16.5	12.4	5.8	-1.6	-4.8	(31)
(32)			MCDWB	-10.2	-6.4	0.7	6.5	14.4	25.4	25.6	19.9	15.4	6.7	-1.1	-4.3	(32)
(33)		10%	WB	-12.8	-9.5	-2.0	2.2	6.4	16.9	17.6	15.3	10.6	3.8	-3.6	-7.2	(33)
(34)			MCDWB	-12.4	-9.0	-0.7	4.6	9.9	23.4	23.8	18.8	12.9	4.8	-3.0	-6.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.8	9.8	10.6	10.3	8.5	11.3	10.9	9.5	7.4	5.6	7.5	7.6	(35)
(36)			MCDBR	7.9	8.3	8.6	10.3	15.0	15.3	15.4	13.6	11.2	6.6	6.3	6.4	(36)
(37)			MCWBR	7.7	8.0	6.5	6.6	8.1	6.6	6.2	7.1	6.4	5.0	6.3	6.3	(37)
(38)		5% WB	MCDBR	8.1	8.2	8.0	8.6	14.7	13.4	13.3	10.8	9.6	6.4	5.9	6.4	(38)
(39)			MCWBR	7.8	7.9	6.4	5.9	8.1	6.5	6.2	6.5	5.9	4.9	6.0	6.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.227	0.280	0.328	0.349	0.360	0.377	0.430	0.402	0.349	0.297	0.218	0.224	(40)	
(41)		taud	1.893	2.014	1.989	2.046	2.245	2.394	2.218	2.278	2.382	2.213	1.920	1.876	(41)	
(42)		Ebn at Noon	309	604	726	811	846	840	772	759	732	592	313	104	(42)	
(43)		Edh at Noon	31	69	109	129	117	104	121	104	76	56	28	13	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.19	0.94	2.50	4.41	5.29	5.51	4.94	3.37	1.98	0.75	0.28	0.08	(44)	
(45)		RadStd	0.02	0.09	0.17	0.45	0.48	0.45	0.54	0.39	0.29	0.12	0.04	0.01	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only		(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
(47) Regional (2 neighbors)		(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A

Nomenclature: See separate page